

claude-windows / da34752f-c664-4262-9602-f5d4cfcc6db4

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\da34752f-c664-4262-9602-f5d4cfcc6db4\subagents\workflows\wf_774b13f4-334\agent-ad695abadfd37f398.jsonl`
- SHA-256: `796be80c77292e1aad4eac566bc177ca71b733b98fdd2fc105b4b2063e1d8331`
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- project: `wf_774b13f4-334`
- session_id: `da34752f-c664-4262-9602-f5d4cfcc6db4`
```

Transcript

1. user (2026-06-12T14:30:01.830Z)

Repo: C:\Users\avidu\Projects\badminton-highlight-indexer (branch docs/pmf-field-signals).

Review these files (read them in full):

- docs/PMF_FIELD_SIGNALS.md (NEW - hypotheses H1-H7, GREEN/AMBER/RED scoring anchors, pre-registered decision rules PROCEED/PIVOT A-C/FALSIFIED, rollup mechanics, guardrails)
- docs/FIELD_VISIT_ANSWER_SHEET.md (NEW - the per-visit capture form a non-technical field associate fills by hand)
- docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md (EXISTING, lightly edited - the recruit hand-out with the 7-question script and ground rules)

Grounding docs (also read): docs/PMF_MARKET_RESEARCH.md (esp. SS6 risks and pricing anchors), docs/NEXT_STEPS.md item 5, docs/COMMERCIALIZATION.md C13 row.

Context: the owner ratified that PMF validation happens by physically visiting Bangalore badminton academies (10-15 visits by a paid non-technical field associate) and reading signals. The kit's job: make 15 visits produce a pre-registered verdict (proceed/pivot/falsified) rather than motivated reasoning. Product = KhelSutra (rally indexing + highlights from static-camera footage; wedge = messy multi-court BYO footage; serving COGS ~\$0.30-0.60/hr via Gemini).

LENS: internal consistency + grounding. Check every cross-reference and number:

- Do the H-to-Q mappings match the flyer's actual questions verbatim?
- Do the COGS/pricing numbers match PMF_MARKET_RESEARCH.md and the 2026-06-10 quality-table figures cited?
- Do the decision-rule rows reference real docs/sections (UI_ALPHA_EXECUTION_PLAN Phase R, PMF SS6 risk numbers, the PB Vision x PodPlay reference)?
- Does anything contradict the repo guardrails (minors exclusion, consent, no-promises) or the flyer's ground rules?
- Markdown/link correctness (relative links resolve, tables well-formed).

2. user (2026-06-12T14:30:09.557Z)

docs/pmf-field-signals

M docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md

M docs/NEXT_STEPS.md

?? arial.ttf

?? artifacts/

?? docs/FIELD_VISIT_ANSWER_SHEET.md

?? docs/PMF_FIELD_SIGNALS.md

?? scratch/HIGH_RISK_REMEDIATION_PLAN.md

?? scratch/PR1_BODY.md

?? scratch/PR1_HANDOFF.md

?? scratch/PR2_HANDOFF.md

?? scratch/PR2_deferred_hardening_status.patch

?? scratch/PR88_FIX.md

3. user (2026-06-12T14:30:14.280Z)

```

1      # C13 Field Signals ? what academy visits must tell us (and when to change course)
2
3      **Status:** ACTIVE ? the decision layer for the C13 beachhead validation
4      (owner-ratified direction, 2026-06-12: *PMF validation is handled by physically
5      visiting academies and reading signals on the ground*).
6      **Companions:** [`FIELD_RESEARCH_ASSOCIATE_FLYER.md`](FIELD_RESEARCH_ASSOCIATE_FLYER.md)
7      (the recruit hand-out: script + ground rules) .
8      [`FIELD_VISIT_ANSWER_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) (the per-visit capture
form) .
9      [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) (the hypotheses under test, esp. §6).
10
11     **Why this doc exists:** the flyer collects answers; this doc fixes ? **before the
12     first visit** ? what those answers *mean* and what totals trigger which decision.
13     Pre-registering the thresholds is the honesty mechanism: it stops us from reading
14     15 ambiguous conversations as whatever we hoped to hear.
15
16     ---
17
18     ## 1. The hypotheses under test (each maps to script questions)
19
20     | ID | Hypothesis (from PMF research) | Falsified looks like | Script Q |
21     |---|---|---|---|
22     | H1 | **Recording behavior exists** ? academies already produce fixed-angle / CCTV /
multi-court footage without us | nobody records; "we'd have to start for you" | Q1 |
23     | H2 | **Dead-footage pain** ? recorded footage piles up unwatched (the indexing wedge)
| footage is small, or already gets watched | Q2 ? |
24     | H3 | **An audience exists** ? coach, player, or parent would actually consume rally
clips/reels | shrug; "who has time to watch clips" | Q3 |
25     | H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory, Stupa, BadPro+) leave
a gap we fit | "we use X, it's fine, fully solved" | Q4 |
26     | H5 | **Willingness to pay per match** at a price that clears unit economics | "should
be free" / <?100 everywhere | Q5 ? |
27     | H6 | **Multi-court chaos is a felt problem** (our moat: indexing messy BY0 multi-court
footage) | multi-court video already useful to them | Q6 |
28     | H7 | **Behavioral commitment** ? they'll let us record on their court (Phase-2 door) |
polite no across the board | Q7 |
29
30     Unit-economics floor behind H5: serving COGS on the Gemini path is ~$0.30?0.60 per
31     hour of video (? 25?50/hr; 2026-06-10 quality table). A 2?3 hr session costs
32     roughly 75?150 to process, so **?300/match has real margin, ?100/match is
33     break-even-ish at retail** ? that's why the script's anchor ladder is ?100/300/500.
34
35     ## 2. Per-visit scoring ? GREEN / AMBER / RED anchors
36
37     Score each hypothesis on the answer sheet **from what they said and showed, not
38     what you inferred**. When between two grades, take the lower one.
39
40     | | GREEN (strong confirm) | AMBER (weak/mixed) | RED (disconfirm) |
41     |---|---|---|---|
42     | **H1 record** | shows or names actual setup (tripod/CCTV/phone rig) used ?monthly |
records occasionally (tournaments only) | doesn't record; no plans |
43     | **H2 dead footage** | points at real unwatched volume (?~5 hrs, "CCTV runs all day",
tournament backlog) | some footage, partially watched | no backlog exists |
44     | **H3 audience** | names WHO would watch and WHEN (e.g. "parents ask for clips", "I
review with students Mondays") | "could be useful, maybe" | no consumer identified |
45     | **H4 whitespace** | no tool today, or uses one and volunteers a gap we cover (multi-
court, BY0 footage, price) | uses a competitor, mildly satisfied | competitor solves it,
satisfied, sticky |
46     | **H5 WTP** | volunteers ??300/match unprompted, or proposes a budget framing of their
own (per-student/per-month) | lands at ~?100 with hesitation | refuses to name a number / "free"
|
47     | **H6 multi-court** | complains about it in their own words before/without prompting |
agrees when asked | multi-court footage already works for them |
48     | **H7 Phase-2** | yes + gives a name/number/slot | "maybe, come back" | no |
49

```

50 **Weighting rule (behavior > words):** H7-GREEN and H2-GREEN are *behavioral*

51 signals and count double in your gut-read; H3/H5 answers are *talk* until a Phase-2

52 session or a real payment corroborates them. A visit with H5-GREEN but H7-RED is a

53 polite conversation, not a customer.

54

55 **Disconfirming-evidence quota:** every visit sheet must record at least one

56 verbatim *negative* quote (the thing they liked least / wouldn't pay for). If a

57 sheet has only positives, the visit gets re-done or discounted ? per the flyer's

58 ground rule 4, a reasoned "no" is worth more than five vague yeses.

59

60 ## 3. Pre-registered decision rules (n = 10?15 academies)

61

62 Counts, not percentages ? at this sample size we're reading direction, not

63 estimating proportions. Evaluate after **every 5 completed sheets** ; final call at

64 the agreed n. "GREEN" below means GREEN on that hypothesis row.

65

66 | Verdict | Trigger (pre-registered) | What we then do |

67 |---|---|---|

68 | **PROCEED ? beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3

H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha

Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the

single warmest |

69 | **PIVOT A ? "index the backlog" ? "record + index" service** | H2 mostly RED (<3

GREEN) but H3+H5 healthy on *fresh* recordings | The pain isn't the archive, it's that nothing

useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the

product wedge, not the demo |

70 | **PIVOT B ? partner/API motion** | H4 RED at ?5 academies (incumbents entrenched and

liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever

owns the academy relationship (the PB Vision x PodPlay template from the 2026-06-10 audit) |

71 | **PIVOT C ? payer is the parent, not the academy** | H5 RED from owners but H3

repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel

to parents) with the academy as channel, not customer |

72 | **FALSIFIED ? beachhead assumption fails** | <3 academies record at all (H1) **and** 0

H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this

market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building

further |

73

74 Mixed outcomes that match no row: write up the per-hypothesis tallies and decide

75 explicitly with the owner ? but **the tallies above get computed first**, so the

76 discussion starts from the pre-registered read, not from anecdotes.

77

78 **Competitor-intel side-channel (always-on):** any visit where a competitor is

79 named ? fill the answer sheet's competitor block (what it does, what it misses,

80 what they pay). ?3 consistent reports about one product = update

81 `PMF\_MARKET\_RESEARCH.md` §4 with field-verified rows (today's table is web-sourced).

82

83 ## 4. Mechanics

84

85 - One [`FIELD\_VISIT\_ANSWER\_SHEET.md`](FIELD_VISIT_ANSWER_SHEET.md) per academy,

86 same day as the visit, in the associate's words + verbatim quotes.

87 - The owner (or Claude session) transcribes each sheet into one row of the rollup

88 table below (keep it in this doc or a sheet next to it); tallies and the

89 decision-rule read-out update per 5 visits.

90 - Demo asset if a conversation earns it: the real compiled reel

91 (`output/MahadevpuraSingles\_highlights.mp4`) on a phone ? *show, never promise*

92 (flyer ground rule 3 still applies; the associate doesn't carry the demo,

93 the owner does on follow-ups).

94

95 ### Rollup table (one row per completed visit)

96

97 | # | Academy | Area | Courts | Role spoken to | H1 | H2 | H3 | H4 | H5 (?) | H6 | H7 |

Gut | Phase-2? |

98 |---|---|---|---|---|---|---|---|---|---|---|---|

99 | 1 | | | | | | | | | | | |

100

```

101     ### Tally ? decision read-out (recompute per 5 visits)
102
103     | Metric | Count | Threshold in play |
104     |---|---|---|
105     | H2 GREEN (dead footage) | /n | ?6 ? PROCEED leg; <3 ? PIVOT A leg |
106     | H5 at ?300+ | /n | ?4 ? PROCEED leg |
107     | H7 GREEN (Phase-2 yes + contact) | /n | ?3 ? PROCEED leg; 0 ? FALSIFIED leg |
108     | H4 RED (incumbent satisfied) | /n | ?5 ? PIVOT B |
109     | H1 any recording | /n | <3 ? FALSIFIED leg |
110     | H5 RED + H3 names parents | /n | recurring ? PIVOT C |
111
112     ## 5. Guardrails that bind the field work
113
114     - **Phase 1 records nothing** ? notes only (flyer rule 7).
115     - **Phase 2 consent rules are hard:** written consent from the academy AND adult
116       players; **sessions with minors are never recorded for our library** (training-pool
117       guardrail, CLAUDE.md); the academy is told plainly what the footage is for.
118     - Answer sheets contain personal contact details ? they stay between the associate
119       and the owner; transcribe to the rollup without phone numbers.
120

```

4. user (2026-06-12T14:30:18.508Z)

```

1     # Field Visit Answer Sheet ? one per academy (fill the same day)
2
3     > Print one per planned visit. Write THEIR words, not yours. A clear "no, because?"
4     > is a *good* sheet. Companion script: `FIELD_RESEARCH_ASSOCIATE_FLYER.md`.
5     > Scoring anchors (G/A/R): `PMF_FIELD_SIGNALS.md` §2.
6
7     ## Visit basics
8
9     | | |
10    |---|---|
11    | Date / time | |
12    | Academy name | |
13    | Area (neighbourhood) | |
14    | # courts in the hall | |
15    | Spoke to (name + role) | |
16    | Contact (phone/WhatsApp) ? *only if offered* | |
17    | How long the conversation ran | |
18
19    ## The seven questions ? their answers (verbatim where possible)
20
21    **Q1 ? Do they record sessions/matches today? How (phone / tripod / CCTV / hired)?**
22
23    >
24
25    **Q2 ? ? Footage sitting around that nobody watches? Roughly how much, and from what?**
26
27    >
28
29    **Q3 ? Who would watch rally clips/highlights (coach / players / parents)? When?**
30
31    >
32
33    **Q4 ? What do they use today (Machaxi / VISIST / Game Theory / BadPro+ / none)?
34    What's good, what's missing, what does it cost them?**
35
36    >
37
38    **Q5 ? ? Would they pay per match for rally cuts + reel + report? What number did
39    THEY say first (?)? What was their face doing?**
40
41    >
42

```

```

43      **Q6 ? Multiple courts in frame ? does that make video useless today? Their words:**
44
45      >
46
47      **Q7 ? Open to a paid recording session on their court? (yes / maybe / no ? and who
48      to call)**
49
50      >
51
52      ## Required: the most negative thing they said (verbatim)
53
54      >
55
56      ## Best quotes (anything vivid, in their words)
57
58      >
59
60      ## Scores (circle one per row ? anchors in PMF_FIELD_SIGNALS.md §2; when torn, score
61      lower)
62      | Hypothesis | Score |
63      |---|---|
64      | H1 recording behavior | GREEN / AMBER / RED |
65      | H2 dead footage | GREEN / AMBER / RED |
66      | H3 audience | GREEN / AMBER / RED |
67      | H4 whitespace | GREEN / AMBER / RED |
68      | H5 willingness to pay ? their ? number: _____ | GREEN / AMBER / RED |
69      | H6 multi-court pain | GREEN / AMBER / RED |
70      | H7 Phase-2 door | GREEN / AMBER / RED |
71
72      ## Competitor block (only if they named one)
73
74      | | |
75      |---|---|
76      | Product | |
77      | What it does well (their words) | |
78      | What it misses (their words) | |
79      | What they pay | |
80
81      ## Your gut read (circle): COLD · POLITE · CURIOUS · EAGER
82
83      ## Phase-2 candidate? (circle): YES ? call: _____ · MAYBE · NO
84
85      ## Anything else worth knowing (layout, lighting, camera positions you noticed, who
86      really decides)
87
88      >

```

5. user (2026-06-12T14:30:18.628Z)

```

1      # Temporary Engagement: Field Research Associate (Sports-Tech, Bengaluru)
2
3      **Duration:** ~2?3 weeks, part-time (flexible hours, mostly afternoons/evenings when
4      academies are active)
5      **Location:** Bengaluru ? in-person visits to badminton academies
6      **Compensation:** ? ___ per completed academy visit + travel reimbursement *(discussed
7      individually)*
8      **Reports to:** Avi (phone/WhatsApp: ___ )
9
10     ---
11
12     ## The gig in one paragraph
13
14     We're building an AI tool that takes ordinary practice/match video from a badminton

```

academy and automatically finds
13 every rally, builds highlight reels, and generates match reports. Before building
further, we need honest, on-the-ground
14 answers from real academies about how they record, what they'd use, and what they'd pay.
**Your job: visit 10?15 good
15 academies, have a friendly 15?20 minute conversation with the owner/head coach using the
script below, and write down
16 what they actually say.** No selling. No tech knowledge needed. If a conversation goes
well, there's a Phase-2 follow-up
17 (coordinating a paid video-recording session) with additional compensation.
18
19 ---
20
21 ## Phase 1 ? The conversation (what you actually do)
22
23 **Who to talk to:** the academy owner or head coach ? the person who decides what the
academy spends money on.
24 If you get a junior coach, be friendly, gather what you can, and ask when the owner is
available.
25
26 **Opening (keep it this honest):**
27 > "Hi, I'm helping a Bengaluru engineer who's building an AI tool for badminton
academies ? it takes normal practice
28 > video and automatically cuts it into rallies and highlights. He asked me to talk to a
few serious academies to
29 > understand how you work with video today. Could I ask you a few questions? Takes 15
minutes, and we're not selling
30 > anything today."
31
32 **The questions (ask in this order; the starred ones matter most):**
33
34 1. Do you ever video-record sessions or matches here? How ? phone, tripod, CCTV, a hired
person?
35 2. ? **Do you have recorded footage sitting around that nobody has time to watch?**
(multi-court hall recordings,
36 tournament days, CCTV) ? roughly how much?
37 3. Who would watch rally clips or highlights if they existed ? coaches for training?
players? parents?
38 4. What do you use today for any of this? (Listen for: Machaxi, VISIST, Game Theory,
BadPro+, just YouTube/phone.)
39 What do you like / what's missing in it?
40 5. ? **If you could hand over a full session's video and get back every rally cut out +
a highlight reel + a simple
41 report ? would you pay per match for that? What feels fair ? ?100, ?300, ?500 per
match?** (Let THEM say a number
42 first. Note their face, not just the words.)
43 6. Your hall has multiple courts running at once, right? Does that make video less
useful today?
44 7. Would you let us record one session here (we pay a small fee for the time) to show
you what the tool produces on
45 YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*
46
47 **After each visit, fill ONE answer sheet ? the template is
FIELD_VISIT_ANSWER_SHEET.md
48 (print one per visit):** academy name, location, # courts, person spoken to + role +
contact, answers 1?7 (their
49 words, not yours), the most negative thing they said (required), exact quotes, your gut
read, and whether they're a
50 Phase-2 candidate. *(How the answers get scored and what totals trigger which decision
lives in
51 PMF_FIELD_SIGNALS.md ? that part is ours, not yours; just
capture honestly.)*
52
53 ---
54

```

55     ## Ground rules & gotchas (read twice ? this is the difference between useful and
useless)
56
57     1. **You are researching, not selling.** If they ask "how much will it cost", say
"that's exactly what he's trying to
58         figure out ? what would feel fair to you?" Never quote a price yourself.
59     2. **Past behavior beats opinions.** "Would you use video?" gets polite lies. "When did
you LAST record a session, and
60         what happened to that file?" gets the truth. Prefer the second kind of question.
61     3. **Don't promise anything.** No features, no dates, no free trials. The only thing you
may offer is the Phase-2 idea
62         (a paid recording session + free sample output) and only as "would you be open to
it".
63     4. **Write down disappointing answers verbatim.** A "no, we'd never pay for that" with a
reason is worth MORE to us
64         than five vague yeses. You're paid per completed honest visit, not per positive
answer.
65     5. **If they already use a competitor (Machaxi/VISIST/Game Theory etc.) ? goldmine.**
Ask what it does well, what it
66         doesn't, and what they pay. Spend extra time here.
67     6. **Multi-court is our favorite topic.** If they complain that video is useless because
"all the courts are in frame
68         and it's chaos" ? capture that complaint in their exact words. That's the problem we
solve.
69     7. **No recordings of people without clear permission.** In Phase 1 you record NOTHING ?
notes only. Don't photograph
70         the academy or players unless the owner offers.
71     8. **Don't oversit.** 20 minutes, thank them, leave a good impression ? we may come back
to record.
72     9. **Confidentiality:** don't describe the tech beyond the opening line, don't name
other academies' answers, and
73         don't share the answer sheets with anyone except Avi.
74
75     ---
76
77     ## Phase 2 (only for academies that said yes ? separate briefing & pay)
78
79     You coordinate a recording session: we pay the academy a small fee, set up a
phone/camera on a tripod (we provide a
80         one-page capture checklist: behind the court, elevated, whole court in frame, landscape,
1080p/60fps, don't move it),
81         record 1?2 sessions, and later return a sample highlight reel + report to the academy
for free. **Hard rules:**
82         written consent from the academy AND the adult players being recorded; **sessions with
minors are not recorded for
83         our library** ? adult sessions only; the academy gets told plainly what the footage is
for (building and demonstrating
84         the tool).
85
86     ---
87
88     ## Interested?
89
90     Message Avi at ___ with "FIELD RESEARCH" and your availability. First 2 visits are
paired (you + Avi or a phone
91         walkthrough) so you're never improvising.
92

```

6. assistant (2026-06-12T14:30:25.362Z)

Now let me read the grounding docs.

7. user (2026-06-12T14:30:26.314Z)

```

1     # PMF & Market Study ? AI Highlight-Reel Tool for Self-Recorded Racket Sports

```

```

2
3 > **Research snapshot ? 2026-06-07.** Figures are cited as-of with source + year;
4 > validate live before relying on them for investment/pricing decisions. This is the
5 > market-research-of-record behind the PMF section of the living
6 > [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) tracker ($6). Product framing:
7 > a closed-source hosted API/SaaS (plus local mode) that ingests long-form,
8 > self-recorded, **static-camera, full-court amateur** racket-sports video and
9 > auto-detects rallies and compiles highlight reels. Primary sport: badminton;
10 > sport-generic architecture roadmapped to tennis, pickleball, table tennis, cricket.
11
12 > **Confidence convention.** Figures from named industry reports, federations, or
13 > company pricing pages are tagged with source + year. Inferred items are tagged
14 > ** (estimate) ** / ** (low confidence) **. Emerging-category market sizes vary widely
15 > between vendors; ranges are shown rather than false precision.
16
17 ---
18
19 ## 1. Executive summary
20
21 The product sits at the intersection of three favorable trends: (a) a fast-growing
22 **sports analytics market** (~USD 5?6B in 2025, ~18?22% CAGR to USD 24?36B by ~2034
23 depending on vendor); (b) a structural shift to **amateur self-recording** on action
24 cameras and phones (action-camera market ~USD 6.5B in 2024, sports the largest
25 segment); and (c) **strong racket-sport participation** ? 220M+ badminton players,
26 106M tennis players (2024), and a near-vertical pickleball curve (~19.8M US players in
27 2024, +45.8% YoY).
28
29 The central tension is the **static-camera constraint**: the tool works best on fixed,
30 full-court amateur footage and *fails* on broadcast/panning footage. This is both a
31 moat (most competitors target team sports / broadcast) and a filter on the addressable
32 user ? the customer must already set a camera on a tripod behind the court. That
33 behavior is concentrated in **coaches/academies, clubs, and serious enthusiasts**, not
34 the median casual player.
35
36 **Recommended beachhead (strongest single):** **badminton coaching academies and
37 serious club/competitive players in India** ? a large, rapidly formalizing academy
38 ecosystem (Khelo India tailwind), genuine fixed-angle recording behavior, existing
39 willingness to pay for coaching, an underserved tooling gap (incumbents are
40 team-sports/broadcast), and a direct-but-immature competitor field. The product is
41 *native badminton-first* ? a genuine edge here. Secondary: ** (2) US/global pickleball
42 players & clubs** (capital-rich, fastest-growing, but SwingVision already strong);
43 ** (3) badminton academies/clubs in East/SE Asia + Denmark** (highest density).
44
45 **Pricing** clusters into hardware-bundled auto-filming (Veo, Trace, Pixellot, Spiideo)
46 and software-only phone AI (SwingVision USD 179.99/yr; BadPro+ pay-per-use). A
47 software-only BYO-camera model undercuts the hardware players; realistic anchors
48 ** (estimate): B2C USD 60?180/yr; B2B academy/club USD 300?1,500/yr per location.**
49
50 **Top risks:** (1) amateurs won't reliably set up a tripod ? TAM collapses to
51 coaches/clubs; (2) free/near-free alternatives cap B2C WTP; (3) capture-quality
52 variance is the core failure mode; (4) badminton's strongest markets (India, Indonesia,
53 China) are the most price-sensitive and hardest to bill at Western SaaS prices.
54
55 ---
56
57 ## 2. Market sizing & trends
58
59 ### 2.1 Sports analytics / sports-video market
60 - **2025 size:** ~USD 5.47B (Precedence Research) to ~USD 5.79B (Fortune Business
61 Insights); other vendors USD 5.6?6.0B. *(2025)*
62 - **Growth:** CAGR 15.7%?22.5%. Precedence ? ~USD 29.75B by 2034 (20.63%); Fortune BI
63 ? USD 24.03B by 2032 (22.5%); MarketsandMarkets (narrower) USD 4.75B by 2030 (15.7%);
64 Future Market Insights ? ~USD 36.2B by 2035 (22.1%). *(reports 2024?2025)*
65 - **Video analytics is the leading segment** within sports analytics (Grand View /
66 Fortune BI) ? the directly relevant sub-market.

```


67 - ****Interpretation (estimate):**** the **amateur/club auto-highlight** slice is a small but
68 fast-growing fraction; most analytics revenue today is pro/broadcast/team-sports. The
69 product targets the long tail the big numbers under-serve.
70

71 **### 2.2 Action-camera / amateur recording adoption**
72 - ****2024 size:**** ~USD 6.5B (GMIInsights) ? ~USD 16.9B by 2034 (~10% CAGR). **(2024?2025)**
73 - ****Sports is the largest end-use segment**** of action cameras (GMIInsights, 2024).
74 - ****GoPro ~47% share, DJI ~20%**** (electroIQ/industry, 2024); phones + cheap tripods are
75 the de-facto amateur rig.
76 - ****Implication:**** the capture-side behavior the product depends on is already
77 widespread and growing ? **for the segments that record** ? validating "BYO-camera,
78 software-only."
79

80 **### 2.3 Participation & popularity by sport and geography**
81

82	Sport	Participation signal	Source (year)	
83	---	---	---	
84	**Badminton**	220M+ play regularly;	160+ countries	BWF est. via BadmintonBites (2023)
85	Badminton ? China	~80M registered,	up to ~100M (largest base)	industry compilations (2023)
86	Badminton ? India	20M+ recreational by 2023;	academy boom + Khelo India	industry compilations (2023)
87	Badminton ? Indonesia	National sport;	~1M club players	industry (est.)
88	Badminton ? Denmark	100,000+ registered club members	(Europe #1)	industry (2023)
89	Badminton ? Japan/Korea	Major school-club + workplace sport		industry (2023)
90	**Tennis**	**106M players (2024)** ,	+25.6% over 5 yrs (87M in 2021)	ITF Global Tennis Report (2021, 2024)
91	Tennis ? Asia	~1/3 of players (33M of 87M)		ITF (2021)
92	**Pickleball ? US**	**19.8M players (2024)** ,	+45.8% YoY, +311% over 3 yrs;	~6.2M core SFIA / Pickleheads / The Dink (2024)
93	Pickleball ? US courts	~70,000 courts,	+55% places to play YoY	SFIA / Pickleheads (2024)
94	Pickleball ? global	IFP 78 member countries;	UK +73% in 2024;	~282M played monthly per PPA claim **(low confidence)** various (2024?2025)
95	**Table tennis**	30M+ players;	227 member associations	ITTF (current)
96	**Cricket**	2.5B+ fans;	500M+ in India (mostly <i>*fans*</i> ,	not self-recorders) Cricket Rise / Nielsen (2024?2025)
97				
98	- **Pickleball market value:**	~USD 2.2B (2024) ? USD 9.1B by 2034 at 15.3% CAGR		
99		(Market.us/Custom Market Insights, 2024?2025).		
100	- **Badminton equipment market:**	~USD 7.23B global (2023),	Asia-Pacific dominant.	
101				
102	**Geographic read:**	badminton density is overwhelmingly Asian + Denmark;	tennis is broad/high-income;	pickleball is US-dominant, capital-rich, globalizing fast;
103				
104				
105				
106				
107				
108				
109	---			
110				
111	## 3. Customer segment evaluation			
112				
113	Segment	Market-size signal	Willingness to pay	Accessibility Static-camera fit B2C/B2B Verdict
114	---	---	---	---
115	**Individual amateur/enthusiast**	Largest raw TAM (100s of millions)	**Low** ? competes with free;	~USD 5?15/mo ceiling High via app stores; high CAC/churn **Mixed** ? most won't set a tripod B2C Volume play, weak economics; use as funnel
116	**Coaches & coaching academies**	India academy boom;	thousands across Asia	**High** ? already sell coaching; USD 300?1,500/yr plausible Reachable via federations/distributors/social **Strong** ? already film fixed-angle B2B/B2B2C **Best fit** ? buyer, recorder, payer aligned
117	**Clubs & leagues**	Denmark 100k+;	thousands of clubs Asia/EU	Medium-high; USD

500?3,000/yr | Medium ? committee sales cycle | ****Strong**** | B2B/B2B2C | Strong secondary; longer cycle |

118 | ****Tournament/event organizers**** | Many small/regional events | Medium ? per-event/seasonal | Medium ? episodic | ****Strong**** | B2B | Good per-event upsell; lumpy |

119 | ****Schools & universities**** | Large (esp. Asia/Japan clubs) | Low-medium ? budget-constrained | Low ? slow procurement | Strong | B2B | Slow; grant/Khelo-India dependent |

120 | ****National/regional federations**** | Few, high-value | Medium-high; procurement-heavy/political | Low ? long cycles | Strong | B2B | Lighthouse/credibility, not beachhead |

121 | ****Broadcasters / streamers**** | High value per deal | High | Low ? incumbents entrenched | ****Poor ? panning = failure mode**** | B2B | ****Avoid**** ? violates core constraint |

122

123 ****Key dynamics:**** the static-camera constraint is a ****B2B filter**** ? coaches/academies/
124 clubs already record from a fixed angle, so the product fits existing behavior (cleanest
125 PMF). B2C is a large low-WTP, high-CAC funnel best used to feed B2B. Broadcasters
126 explicitly violate the constraint ? exclude.

127

128 ---

129

130 **## 4. Competitive & pricing benchmarks**

131

Product	Category	Model	Price	Notes (year)
---	---	---	---	---
Veo (Cam 3)	Team-sports auto-filming	Hardware + sub	Camera ~USD 1,533 (5G ~1,886) + ~£395?895/yr	15,000+ clubs/80 countries; \$115M raised (2025)
Trace	Soccer auto-filming	**Free camera + sub**	USD 180/yr (Basic) ? 300 (Pro)	Camera leased free with sub (2025)
Pixellot (Air)	Auto camera/streaming	Hardware + sub	Camera ~USD 949 + USD 69?167/mo	Institutional/league; 10,000+ clubs (2025)
Spiideo	Auto-filming/analytics	Sub incl. camera	~USD 3,800/yr (1 team) ? 6,100/yr (club)	Institutional budgets (2025)
Hudl	Team video analysis	SaaS tiers	From ~USD 400/yr	Broad team-sports incumbent (2026)
SwingVision	**Tennis/pickleball AI, single iPhone**	Software freemium	**USD 179.99/yr** ; free 2 hrs/mo	Closest analog; 200k+ monthly users; \$10M+ raised (2025)
BadPro+	**Badminton AI on phone**	**Pay-per-use, no sub**	Per-analysis (? "<one coaching session")	Direct badminton competitor; shot detection, heatmaps, auto-highlights (2025)
	Niche pickleball/badminton apps	Various	Mixed	Varies Fragmented; few true auto-highlight tools (2025)

142

143 ****What buyers actually pay (synthesis):**** consumer software-only anchors at ~USD
144 100?200/yr (SwingVision 179.99); hardware-bundled USD 180?900/yr prosumer/club,
145 USD 3,800?6,100/yr institutional; badminton-specific BadPro+ uses ***pay-per-use*** ?
146 signaling a real badminton AI market but ****unproven subscription WTP**** (opportunity +
147 warning). ****Positioning:**** software-only + BYO-camera undercuts hardware players and
148 sells worldwide via SaaS. Realistic anchors ****(estimate):** B2C USD 60?180/yr; B2B
149 academy/club USD 300?1,500/yr** per location. The differentiators (cheap offline
150 self-improving detection, per-rally stats/tagging, clean licensing) support a ****B2B****
151 value story more than a B2C one.

152

153 ---

154

155 **## 5. Recommended beachhead**

156

157 Ranked by ****WTP × participation density × existing recording behavior × static-camera**
158 **fit × competitive whitespace.****

159

160 **### #1 (strongest) ? Badminton coaching academies & serious players, ****India******
161 20M+ recreational players (2023); rapidly ****formalizing academy ecosystem**** + Khelo
162 India tailwind; academies already record fixed-angle; coaches are buyers who monetize
163 instruction (high relative WTP); incumbents ignore badminton; the only direct badminton
164 AI (BadPro+) is early with unproven pay-per-use. Product is ***native badminton-first***.
165 ****Watch-outs:**** price sensitivity (need INR-tuned tiers); sell to the ****academy****, not
166 the individual.

167

168 ### #2 ? Serious pickleball players & clubs, **US (then UK/Australia/Asia)**
169 Fastest-growing sport (19.8M US, +45.8% YoY, 2024); capital-rich; ~70,000 courts;
170 strong recording/sharing culture; pickleball is on the roadmap. **Watch-out:**
171 **SwingVision already owns single-camera tennis+pickleball AI** at USD 179.99/yr (200k+
172 monthly users) ? head-to-head differentiation, not greenfield.
173
174 ### #3 ? Badminton academies/clubs, **East/SE Asia (China, Indonesia, Malaysia, Japan,
Korea) + Denmark**
175 Highest global density (China 80M+; Denmark 100k+ structured club members; Japan/Korea
176 school-club & workplace systems), mature club infra, fixed-court recording.
177 **Watch-outs:** China needs local hosting/partners + is regulatory-complex; SE Asia
178 price-sensitive; Denmark/EU small but high-WTP and easy to pilot.
179
180 **Bottom line:** lead with **India badminton academies** (whitespace + behavior + buyer
181 alignment), use **enthusiast B2C as the funnel**, validate a **club/academy B2B2C
182 motion**, then expand to **pickleball-US** (out-build SwingVision) and
183 **badminton-Asia/Denmark** (density).
184
185 ---
186
187 ## 6. Risks & open questions (what would invalidate the thesis)
188
189 1. **Tripod/static-camera adoption among amateurs.** If serious amateurs won't set up a
190 fixed camera, B2C TAM collapses to coaches/clubs. *Validate:* observe 3?5 academies +
191 50?100 enthusiasts on whether they already record fixed-angle.
192 2. **Free / near-free alternatives** (YouTube, free app tiers, BadPro+ pay-per-use) cap
193 B2C WTP. *Validate:* fake-door pricing at USD 60?180/yr.
194 3. **Capture-quality variance = the failure mode** (gym lighting, off-center angle,
195 occlusion, mixed resolutions). *Validate:* benchmark rally-detection P/R on 100+ real
196 amateur clips; define a "good-capture" spec + auto-reject (ties to
197 `VIDEO\_RECORDING\_GUIDELINES.md` + the observability reports).
198 4. **Monetization mismatch in best-participation markets** (India/Indonesia/China are
199 price-sensitive, hard to bill). *Validate:* local pricing/payments pilot; per-academy
200 INR licensing.
201 5. **Competitive response** (SwingVision or a team-sports incumbent adds racket sports).
202 *Validate:* ship a badminton-quality advantage incumbents can't trivially match
203 (offline self-improving detection, per-rally tagging).
204 6. **B2B sales-cycle drag** for clubs/federations/schools. *Validate:* single-coach/
205 academy self-serve before institutional deals.
206 7. **Clean-licensing positioning** matters mainly to B2B buyers ? confirm it's a real
207 purchasing criterion via 5?10 buyer interviews.
208
209 **Cheapest next steps:** (a) 10?15 discovery interviews with Indian badminton academy
210 owners/coaches; (b) a capture-quality benchmark on real amateur footage; (c) a fake-door
211 pricing test across B2C (~USD 99/yr) and B2B academy (~USD 600/yr).
212
213 ---
214
215 ## 7. Sources
216
217 - Sports analytics market: Precedence Research; Fortune Business Insights; Grand View
218 Research; MarketsandMarkets; Future Market Insights. *(2024?2025)*
219 - <https://www.precedenceresearch.com/sports-analytics-market>
220 - <https://www.fortunebusinessinsights.com/sports-analytics-market-102217>
221 - <https://www.grandviewresearch.com/industry-analysis/sports-analytics-market>
222 - <https://www.marketsandmarkets.com/PressReleases/sports-analytics.asp>
223 - <https://www.futuremarketinsights.com/reports/sports-analytics-market>
224 - Action camera market: [https://www.gminsights.com/industry-analysis/action-camera-](https://www.gminsights.com/industry-analysis/action-camera-market)
market (2024);
225 <https://www.precedenceresearch.com/action-camera-market> (2024);
226 <https://electroi.com/stats/action-camera-statistics/> (2024)
227 - Badminton participation/market: [https://badmintonbites.com/how-many-badminton-players-](https://badmintonbites.com/how-many-badminton-players-are-there-in-the-world/)
are-there-in-the-world/ ;
228 [https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-](https://stringsandpaddles.com/where-is-badminton-played-10-countries-where-its-popular/)
popular/ ;

229 <https://en.wikipedia.org/wiki/Badminton> ;
 230 <https://www.marketgrowthreports.com/market-reports/badminton-market-110049> ;
 231 <https://www.thesportsschool.com/top-10-badminton-academies-in-india/>
 232 - Tennis: <https://www.itftennis.com/en/news-and-media/articles/itf-global-tennis-report-participation-hits-106-million-in-five-years/> ;
 233 <https://sgbonline.com/itf-global-tennis-report-2021-shows-worldwide-participation-gains/>
 234 - Pickleball: <https://sfia.org/resources/as-pickleball-continues-unprecedented-growth-in-every-age-group-and-region-for-third-straight-year-significant-investments-still-needed-for-court-and-facility-demand/> ;
 235 <https://www.thedinkpickleball.com/report-pickleball-participation-surges-to-nearly-20-million-in-2024/> ;
 236 <https://www.pickleheads.com/guides/pickleball-statistics> ;
 237 <https://www.custommarketinsights.com/report/pickleball-market/> ;
<https://market.us/report/pickleball-market/>
 238 - Table tennis: https://en.wikipedia.org/wiki/International_Table_Tennis_Federation
 239 - Cricket: <https://cricketrise.com/cricket-fans-in-india/> ;
 240 <https://www.nielsen.com/insights/2025/global-cricket-connecting-fans-driving-roi/>
 241 - Competitors & pricing: <https://www.véo.com/en-us/pricing> ; <https://traceup.com/pricing>
 ;
 242 <https://www.pixellot.tv/> ; <https://www.hudl.com/pricing> ;
<https://swing.vision/subscribe> ;
 243 <https://www.badproplusapp.com/>
 244

8. user (2026-06-12T14:30:26.615Z)

```

1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha
track opened**:
4      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
5      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
6      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
7      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
8      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
9      `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
10     **Authoritative roadmap:**
11     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
12     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
13
14     ## Where we are (one paragraph)
15     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
16     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
17     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
18     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
19     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
20     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
21     battery is done** (see table) ? the moat is quantified.
22
23     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)

```

```

24 | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
25 |---|---|---|
26 | Heuristic (velocity windowing) | 0.657 | 0.157 |
27 | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
28 | Two-stage WASB+Gemini refine | 0.83 | ? |
29 | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
30
31 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
32 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
33 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
34 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
35 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
36 Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
37
38 ## Prioritized next steps
39 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
every lever below feeds on it.
40 Priority order for new videos: **(a) multi-court badminton** (the target distribution
AND where the ship target
41 lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
0.909; pickleball labels are
42 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
capture variety per
43 `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
(consent guardrail). Per video:
44 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
re-run the Gen-0 harness.
45 The paired sign test needs n: at n=10 with 9 wins, p>0.011 ? significance is
reachable this month.
46 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
(necessary), **multi-court
47 reference = wrapper 0.66** (the number that makes the owned model worth serving),
clean-footage ceiling = 0.93
48 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70
on multi-court folds with
49 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
50 3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for
**ActionFormer/ASFormer** (MIT, minutes on the
51 2070S ? still lightweight, NOT the heavyweight Gen-2 stack) inside the existing
harness; per-video threshold
52 calibration (the LOVO-vs-oracle gap is visible per fold in the harness output).
53 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
merged-prototype LOVO across
54 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-
on-WASB serving stays gated
55 by C3 (provenance multiplies per sport).
56 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+
the two added questions:
57 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
58 (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
59 `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
60 physical-visit approach):** `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (recruit hand-
out: script, ground rules,
61 Phase-2 consent; owner fills ?/contact placeholders) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit capture
62 form) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R scoring anchors + **pre-registered

```

```

decision rules** ? PROCEED /
63     PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
64     6. **Housekeeping:** ~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
65     deletion in AI Studio); **rename repo off "badminton"*** (owner action,
66     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
67
68     ## Pointers / artifacts
69     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
70     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
71     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
72     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
73     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
74     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
75     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
76     labels: `...\Collect\Golden Labelled\`.
77     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
78     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
79     with a MISLEADING "credits depleted" 429; ramp gradually.
80     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
81     PLATFORM slice = async job model (single-tenant).
82

```

9. user (2026-06-12T14:30:32.668Z)

```

32-| # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
33-|---|---|---|---|---|---|
34-| | **? PENDING (needs action / not started)** | | | |
35-| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC**
(`libx264` in `backend\pipeline\stitcher\compiler.py`) ? encode-side AVC patent exposure
remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4\L40)** for
HW encode. See $4. Decision: provision Ada-class GPU? |
36-| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37-| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks
on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md`
(pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md`
(#16). |
38-| | **? IN-FLIGHT (in progress / watch)** | | | |
39-| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **(C) Gemini 2.5
Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open
VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make
"provider registry + paid fallback" a core principle; build the owned model **after** the
scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` $5A. |
40-| C13 | **Beachhead validation** | **P0** | ? | Market research (`PMF_MARKET_RESEARCH.md`)
recommends **India badminton academies** as the #1 beachhead. Unvalidated. | Cheapest tests:
10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur
clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See $6. |
41-| C8 | **Golden-set vendoring spend** | **P1** | ? | ADR-006: **Roora (Bengaluru)** selected
for an initial paid pilot; licensed/owned video only (no user uploads to vendors). | Confirm
quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See
`GOLDEN_SET_VENDORING.md`. |
42-[Omitted long context line]
43-[Omitted long context line]
44-| C6 | **Inbound HEVC decode** | **P2** | ? ?? | Decoding users' GoPro H.264/**HEVC** uploads
is unavoidable; HEVC pools are fragmented + highest-uncertainty. | Low/uncertain residual;
confirm decode-side exposure with counsel before launch. |

```

45-| C7 | ****Hosting unit-economics**** | ****P2**** | ? | Per-video cost is **cheap** for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | ****Benchmark \$/video & latency on the target VM**** for A/B/C before pricing. See §5. |

46-| C10 | ****Value layer (product surface)**** | ****P2**** | ? | Minimal exports shipped (EDL\JSON\CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6. |

47-| C11 | ****Input-quality enforcement**** | ****P2**** | ? | Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES` dimensions into validators/report rules. |

48-| C2 | ****`supervision` ByteTrack pin**** | ****P2**** | ? | Pinned `<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one. |

--

196- throughput (C7).

197-5. ****Provision Ada-class GPU (L4/L40)**** for HW AV1 encode? (C5 ? gates the codec switch.)

198-

199----

200-

201-## Changelog

202-

203-- ****2026-06-07**** ? Added the market-research-of-record (`PMF\_MARKET\_RESEARCH.md`):

204: ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13

205- "beachhead validation" workstream. Wired the summary into §6.

206-- ****2026-06-07**** ? Created this living tracker. Merged the 2026-05-30 monetization

207- audit + the commercial-readiness review. ****Refreshed since the audit:**** C1 AGPL

208- removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,

209- the audit's AV1 recommendation is not yet implemented); C9 packaging tied to

210- `DEFERRED\_HARDENING\_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer

211- exports). Archived `COMMERCIAL\_READINESS\_REVIEW.md`.

212-

10. user (2026-06-12T14:30:34.035Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-65-

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-66-| Verdict | Trigger (pre-registered) | What we then do |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-67-|---|---|---|

docs\PMF_FIELD_SIGNALS.md:68:| ****PROCEED ? beachhead validated**** | ?6 H2-GREEN ****and**** ?4 H5 at ?300+ ****and**** ?3 H7-GREEN | Run the camera pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list (UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-69-| ****PIVOT A ? "index the backlog" ? "record + index" service**** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on **fresh** recordings | The pain isn't the archive, it's that nothing useful gets made at all ? lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |

docs\PMF_FIELD_SIGNALS.md:70:| ****PIVOT B ? partner/API motion**** | H4 RED at ?5 academies (incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the engine/API route through whoever owns the academy relationship (the PB Vision × PodPlay template from the 2026-06-10 audit) |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-71-| ****PIVOT C ? payer is the parent, not the academy**** | H5 RED from owners but H3 repeatedly names parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with the academy as channel, not customer |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-72-| ****FALSIFIED ? beachhead assumption fails**** | <3 academies record at all (H1) ****and**** 0 H7-GREEN | The "academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ? revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-73-

--

docs\UI_ALPHA_EXECUTION_PLAN.md-50-- [] ****D3:**** deployed smoke E2E (owner-only allowlist); go-live security checklist (HOSTING_PLAN §7) all green

docs\UI_ALPHA_EXECUTION_PLAN.md-51-- [] ****GATE GD:**** checklist signed off

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-52-

docs\UI_ALPHA_EXECUTION_PLAN.md:53:### Phase R ? rollout

docs\UI_ALPHA_EXECUTION_PLAN.md-54-- [] Wave 1: 5 testers (C13/camera-pilot contacts) +

feedback channel (WhatsApp group or Google Form)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\UI_ALPHA_EXECUTION_PLAN.md-55-- []
2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL §4)
docs\UI_ALPHA_EXECUTION_PLAN.md-56-- Metrics: matches/week · % rallies corrected · median
turnaround · verdict mix · week-2 tester retention

11. user (2026-06-12T14:32:11.630Z)

C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-67-|---|---|---|
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-68-| **PROCEED ?
beachhead validated** | ?6 H2-GREEN **and** ?4 H5 at ?300+ **and** ?3 H7-GREEN | Run the camera
pilot at the 2?3 warmest academies; those contacts seed the UI-alpha Wave-1 invite list
(UI_ALPHA_EXECUTION_PLAN Phase R); start the paid-pilot conversation with the single warmest |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-69-| **PIVOT A ?
"index the backlog" ? "record + index" service** | H2 mostly RED (<3 GREEN) but H3+H5 healthy on
fresh recordings | The pain isn't the archive, it's that nothing useful gets made at all ?
lead with the Phase-2 recording offer; camera pilot becomes the product wedge, not the demo |
docs\PMF_FIELD_SIGNALS.md:70:| **PIVOT B ? partner/API motion** | H4 RED at ?5 academies
(incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the
engine/API route through whoever owns the academy relationship (the PB Vision × PodPlay template
from the 2026-06-10 audit) |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-71-| **PIVOT C ?
payer is the parent, not the academy** | H5 RED from owners but H3 repeatedly names
parents/players asking for clips | Re-test WTP as B2C per-student reels (?/reel to parents) with
the academy as channel, not customer |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-72-| **FALSIFIED ?
beachhead assumption fails** | <3 academies record at all (H1) **and** 0 H7-GREEN | The
"academies already record fixed-angle" premise (PMF §6 risk 1) is wrong for this market ?
revisit beachheads #2/#3 (pickleball-US, badminton Asia/Denmark) before building further |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-73-

12. user (2026-06-12T14:32:13.395Z)

docs\BACKLOG.md-33-*(source: PR #8, ADR-005)*
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\BACKLOG.md-34-
docs\BACKLOG.md:35:- ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as
of supervision 0.28.0 and **removed in 0.30.0** ; `requirements.txt` is pinned `<0.30.0` as a
stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
docs\BACKLOG.md-36-- ? **Real-video validation of the swap** ? the torchvision + ByteTrack
Stage-1 is unit-tested with mocks but **not yet validated on actual footage** for
detection\tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy
on broadcast video, so expect tuning.) **Revisit:** when a real test clip + the pipeline are run
together.
docs\BACKLOG.md-37-- ? **`yolo\_hybrid` naming** ? the registry key, `yolo\_` config keys, and
`yolo-hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
--
docs\COMMERCIALIZATION.md-46-| C10 | **Value layer (product surface)** | **P2** | ? | Minimal
exports shipped (EDL\JSON\CSV + `basic\_stats`, PR #51). | Deepen: per-rally tags/notes that feed
golden, per-player/match stats, clip gallery. See §6. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-47-| C11 |
Input-quality enforcement | **P2** | ? | Per-video observability reports surface a customer
verdict + footgun flags incl. audio-present (PR #50). | Turn more `VIDEO\_RECORDING\_GUIDELINES`
dimensions into validators/report rules. |
docs\COMMERCIALIZATION.md:48:| C2 | **`supervision` ByteTrack pin** | **P2** | ? | Pinned
`<0.30.0` (deprecation ahead). | Migrate before lifting the pin (BACKLOG). Not a legal risk, a
maintenance one. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-49-| | **?
COMPLETED** | | | |
docs\COMMERCIALIZATION.md-50-| C1 | **AGPL \ copyleft removal** | ? | ? | ADR-005 shipped: AGPL
`ultralitics` YOLOv8 replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack
(MIT)** . No AGPL code/weights in the stack. | Done. Keep new deps permissive. |
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-62-|


```

~~\ultralytics` + `yolov8n.pt`~~ | AGPL-3.0 | Yes ($13) | ? **REMOVED** | Replaced per ADR-005 ?
no longer in the stack. |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-63-|
**torchvision** detectors (Faster R-CNN/RetinaNet) | BSD-3 | No | ? LIVE | The chosen
replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`). |
docs\COMMERCIALIZATION.md:64:| **`supervision`** (ByteTrack) | MIT | No | ? LIVE | Pinned
`<0.30.0` ? migrate before bump (C2). |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-65-| **WASB-SBDT**
(shuttle) | MIT (code) | No | ? code clear | Weights provenance open (C3). |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-66-| **MonoTrack /
YOLO-NAS** | noncommercial | ? | ? BANNED | Do not use. |
--
docs\COMMERCIALIZATION.md-120-Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-121-
docs\COMMERCIALIZATION.md:122:- ** (C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output
$2.50/1M; video ~263
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-123- tok/s ?
10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\COMMERCIALIZATION.md-124- economic
risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md-24- script
? its name collides with setuptools semantics and will confuse `pip`/`build`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md-25-- Deps
live only in `requirements.txt`; `obsreport` is a commented soft dep
docs\DEFERRED_HARDENING_PLAN.md:26: (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md-27-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md-28-###
Approach
--
docs\DEFERRED_HARDENING_PLAN.md-127-- **Detector\WSL abstraction (#6):** in flight (PR #49).
docs\DEFERRED_HARDENING_PLAN.md-128-- **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-
deferred.
docs\DEFERRED_HARDENING_PLAN.md:129:- **ByteTrack `supervision<0.30.0` migration + real-footage
validation (#14/#15):**
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DEFERRED_HARDENING_PLAN.md-130- need a
GPU/WSL box and real video ? owner-machine work, tracked separately.
--
docs\archives\MONETIZATION_AUDIT.md-77-Reference workload: ~30-min 1080p match; processing
happens on a cloud NVIDIA GPU VM.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\MONETIZATION_AUDIT.md-78-
docs\archives\MONETIZATION_AUDIT.md:79:- ** (C) Gemini 2.5 Flash (paid):** input **$0.30 / 1M
tokens** (text/image/video), output **$2.50 / 1M**. Video tokenizes at **~263 tokens/sec** at
default resolution ? a 10-min clip ? 158k input tokens ? **~$0.05 input** (+ small output). Per-
video API cost is **cheap**; the economic risk is *volume × latency*, not unit price. Flash-Lite
($0.10/$0.40 per 1M) is a cheaper tier; Pro is ~10× pricier.
docs\archives\MONETIZATION_AUDIT.md-80-- ** (B) Self-hosted Qwen2.5-VL-7B:** **$0 API**, cost =
GPU-hours amortized over throughput. Needs ?24 GB VRAM (L4/A10-class). Real per-video
cost/latency vs. Gemini is an **open question** ? measure before committing.
docs\archives\MONETIZATION_AUDIT.md-81-- ** (A) Offline ActionFormer:** cheapest at inference
(small model, no LLM, no API); cost is the **one-time training** + light GPU inference. Best
steady-state margin.
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-28-| H7 |
**Behavioral commitment** ? they'll let us record on their court (Phase-2 door) | polite no
across the board | Q7 |
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-29-
docs\PMF_FIELD_SIGNALS.md:30:Unit-economics floor behind H5: serving COGS on the Gemini path is
~$0.30?0.60 per
docs\PMF_FIELD_SIGNALS.md:31:hour of video (? 225?50/hr; 2026-06-10 quality table). A 2?3 hr
session costs
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-32-roughly 75?150
to process, so **?300/match has real margin, ?100/match is
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\PMF_FIELD_SIGNALS.md-33-break-even-ish
at retail** ? that's why the script's anchor ladder is ?100/300/500.

```

--
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-170-- **The learned model has
led since n=3 and holds ~+0.10** (n=6: learned 0.583 vs heuristic 0.480). It stays robust as
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-171- the corpus diversifies;
the heuristic stays capped by the multi-court videos it can't represent (per-fold n=6: Adarsh
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:172: 0.751, doubles 0.737,
Mahadevpura 0.657, but Kushagra **0.157** / gBaaddy **0.309**).
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-173-- **n=6 added
MahadevpuraSingles** (borderline multi-court, contrast 2.2x): held out and trained on the *other
5*, the
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\archives\quality-
iterations\2026-06-multivideo-lovo\README.md-174- learned model hit **AUC 0.902 / F1 0.712**
(vs heuristic 0.657) ? clean generalization to an unseen match.

13. user (2026-06-12T14:32:30.319Z)

docs\FIELD_VISIT_ANSWER_SHEET.md:33:**Q4 ? What do they use today (Machaxi / VISIST / Game
Theory / BadPro+ / none)?
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:38:4. What do you use today for any of this? (Listen for:
Machaxi, VISIST, Game Theory, BadPro+, just YouTube/phone.)
docs\FIELD_RESEARCH_ASSOCIATE_FLYER.md:65:5. **If they already use a competitor
(Machaxi/VISIST/Game Theory etc.) ? goldmine.** Ask what it does well, what it
docs\PMF_FIELD_SIGNALS.md:25:| H4 | **Whitespace** ? incumbents (Machaxi, VISIST, Game Theory,
Stupa, BadPro+) leave a gap we fit | "we use X, it's fine, fully solved" | Q4 |

14. user (2026-06-12T14:32:31.896Z)

c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-385- loss; (9) TT-on-WASB idea silently walks into the C3 ship-
gate ("multiplies per sport"); (10) ship-gate F1@tIoU still
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-386- undefined while 0.44/0.54/0.480 floors circulate; repo-
rename "ASAP" tracked in no tracker (BACKLOG lacks it).
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-387-- **PMF (web-grounded, sources in audit output):**
pricing/cost benchmarks VERIFIED (SwingVision \$179.99/yr ~20k paying
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:388: subs rev \$2.75M +12% YoY; PB Vision \$99/yr + **\$8/hr-of-
video partner API**; Gemini COGS ~\$0.30?0.60/hr ? fat margin);
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-389- but **"India badminton whitespace" FALSIFIED ? Bangalore
is the most contested badminton-AI city** : Machaxi (\$1.5M
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-390- Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru
academies, Padukone School), Game Theory (Gopichand-backed smart
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-391- courts w/ auto-highlights), Stupa (\$3.3M, BWF-approved,
TT?badminton) ? NONE in our 2026-06-07 competitive table.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-392- Surviving wedge (narrow, real): **software-only rally-
precise indexing of EXISTING messy multi-court BYO footage**
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:393: (market solves it only with per-court hardware) sold per-
hour as engine/API, possibly THROUGH those players
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md:394: (PB VisionxPodPlay template). Fastest falsification: C13
interviews (still undone) + **Gemini-wrapper baseline vs the
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-395- n=6 LOVO corpus** (if wrapper ? learned 0.583, owned-model
is premature) + 1 paid academy pilot.
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-396-- **AUDIT NOW COMPLETE (8/8 auditors, resumed `wa4d2980l`
after Max-20x refresh; 40 agents).** New high/critical findings
c:\Users\avidu\claude\projects\c--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md-397- beyond the first batch:

15. user (2026-06-12T14:33:26.562Z)

```
diff --git a/docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md b/docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
index d0d3ba3..1d41402 100644
--- a/docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
+++ b/docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
@@ -44,9 +44,11 @@ If you get a junior coach, be friendly, gather what you can, and ask when the
ow
    7. Would you let us record one session here (we pay a small fee for the time) to show you what
the tool produces on
    YOUR court? *(This is the Phase-2 door ? just note yes/no/maybe and who to call.)*

- **After each visit, fill ONE answer sheet (template on the back / separate page):** academy
name, location, # courts,
- person spoken to + role + contact, answers 1?7 (their words, not yours), any exact quotes, your
gut read (cold /
- curious / eager), and whether they're a Phase-2 candidate.
+ **After each visit, fill ONE answer sheet ? the template is
+ [ `FIELD_VISIT_ANSWER_SHEET.md` ] (FIELD_VISIT_ANSWER_SHEET.md)
+ (print one per visit):** academy name, location, # courts, person spoken to + role + contact,
answers 1?7 (their
+ words, not yours), the most negative thing they said (required), exact quotes, your gut read,
and whether they're a
+ Phase-2 candidate. *(How the answers get scored and what totals trigger which decision lives in
+ [ `PMF_FIELD_SIGNALS.md` ] (PMF_FIELD_SIGNALS.md) ? that part is ours, not yours; just capture
honestly.)*

---

diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index 9032dc7..1d211e5 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -56,8 +56,11 @@ Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2
    5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two
added questions:
    "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the
warmest 2-3 academies
    (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern
exists:
-   `output/MahadevpuraSingles_highlights.mp4`). **Recruiter hand-out ready:
`docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md`**
-   (script, ground rules, Phase-2 consent rules; owner fills ?/contact placeholders).
+   `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
+   physical-visit approach):** `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (recruit hand-out:
script, ground rules,
+   Phase-2 consent; owner fills ?/contact placeholders) + `docs/FIELD_VISIT_ANSWER_SHEET.md`
(per-visit capture
+   form) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R scoring anchors + **pre-registered decision
rules** ? PROCEED /
+   PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
    6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working;
confirm old-key
    deletion in AI Studio); **rename repo off "badminton"** (owner action,
    in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery
completed.
```

16. user (2026-06-12T14:33:26.932Z)

```
docs\BACKLOG.md:17:## Engineering hardening deferred from the 2026-06-10 audit sweep
docs\CODE_MAP.md:7:[Omitted long matching line]
docs\PMF_FIELD_SIGNALS.md:70:| **PIVOT B ? partner/API motion** | H4 RED at ?5 academies
(incumbents entrenched and liked) | Don't fight installed hardware head-on; explore the
```

engine/API route through whoever owns the academy relationship (the PB Vision × PodPlay template from the 2026-06-10 audit) |

17. user (2026-06-12T14:33:50.221Z)

```
=== docs/PMF_FIELD_SIGNALS.md ===
table@L20: header 4 cols
table@L40: header 4 cols
table@L66: header 3 cols
table@L97: header 14 cols
table@L103: header 3 cols
=== docs/FIELD_VISIT_ANSWER_SHEET.md ===
table@L9: header 2 cols
table@L62: header 2 cols
table@L74: header 2 cols
=== docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md ===
```

18. assistant (2026-06-12T14:34:20.093Z)

All tables are well-formed. Let me check the demo asset existence and one last cross-reference detail.

19. user (2026-06-12T14:34:22.148Z)

True

20. user (2026-06-12T14:34:22.156Z)

```
1      # UI Alpha Execution Plan ? KhelSutra Guru
2
3      **Status:** ACTIVE · updated 2026-06-10 · **Linked:** `UI_PROPOSAL.md` ·
`HOSTING_PLAN.md` ·
4      `mockups/ui_alpha_mockup.html` · ADR-009 (`archives/decisions/DECISIONS.md`).
5      **Sequencing principle (owner-requested):** build and test **everything locally first**·
6      deployment (DNS/tunnel/Pages/Access) starts only after the local end-to-end gate passes.
7
8      ## Decisions locked
9
10     | Date | Decision |
11     |---|---|
12     | 2026-06-10 | Product name **KhelSutra Guru** · domain **khelsutra.guru** (GoDaddy,
3-yr + protection) |
13     | 2026-06-10 | Separate **private repo `khelsutra`** ("heart and brain"): `web/` SPA
now, control plane at beta; backend enablers stay in the indexer repo |
14     | 2026-06-10 | **Local-first delivery** (this plan's phase structure) |
15     | open | Final "go" on PR-1 coding start (= #16 greenlight) · training opt-in default
OFF · invite list · stack (React+Vite+TS+Tailwind stands unless owner objects) |
16
17     ## Master checklist
18
19     ### G0 ? Setup (now)
20     - [x] Name/domain/repo-split ratified; spelling confirmed via GoDaddy receipt
21     - [x] `khelsutra` repo **scaffolded** at `C:\Users\avidu\Projects\khelsutra` (README,
.gitignore, web/, docs/)
22     - [x] **Owner:** git-init + private GitHub push ? done 2026-06-10:
`https://github.com/avidullu/khelsutra` (private; git runs Windows-side per ADR-009)
23     - [x] **Owner:** explicit "go" for PR-1 ? given 2026-06-10
24     - [x] **Owner:** Gemini API key rotated ? 2026-06-10, owner-verified (confirm old-key
deletion in AI Studio)
25
26
27     ### Phase L ? build & test locally (no cloud; ?6?7 sessions)
28
29     **L1 ? backend enablers** (indexer repo · branch off master · ONE PR each, no stacking ·
offline tests in every PR · GPU/real-video checks = owner):
```

```

30      - [x] **PR-1 · B1 async job model (#16):** `jobs` table (user_version 1?), `POST
/api/process` ? 202 + `job_id`, `GET /api/jobs/{job_id}`, ThreadPoolExecutor worker,
restart/crash state handling ? **MERGED 2026-06-11, PR #87** (CI green: 333 passed; machine-side
real-video checks fold into L3)
31      - [x] **PR-2 · B2+B3 upload & download:** chunked upload (init ? sequential ?95 MB parts
? complete ? MD5; caps + mp4/mov allowlist; per-user dirs deferred to PR-3) + streaming `GET
/api/highlights/{video_id}/{name}` ? **MERGED 2026-06-11, PR #99** (also fixes static-UI
`segment_ids` compile bug + adds browser download link). **Live E2E verified at merge:** 86 MB
real video uploaded in 3 sequential parts and reassembled to MD5 byte-identity; out-of-order 409
/ traversal 400 (handler) + 404 (router) / unknown-id 404 confirmed; compile of a real confirmed
rally ? streamed ? download (3.9 MB reel).
32      - [ ] **PR-3 · B4+B7 identity & exposure:** `owner_email` column + scoping on every
query; Access-JWT/header read with `DEV_USER_EMAIL` env fallback for local dev; CORS narrowed;
rate limit; per-user quota
33      - [ ] **PR-4 · B6 correction loop:** `PATCH /api/segments/{id}` accept/reject/boundary-
nudge ? `candidate_segments.human_label`/`notes` + provenance (who/when); retire
`backend/api/static` (B5)
34
35      **L2 ? SPA** (khelsutra repo, `web/`; dev server proxies to `http://localhost:8000`):
36      - [ ] **M1 scaffold:** Vite + React + TS + Tailwind, router, typed API client, branding
(KhelSutra Guru)
37      - [ ] **M2:** A2 Upload (chunked, guidelines checklist) + A3 Job Progress (poll)
38      - [ ] **M3:** A1 Matches + A4 Rally Review (player seek, accept/reject/nudge wired to
PR-4)
39      - [ ] **M4:** A5 Highlights (compile + download) + A6 Report + engine-offline banner
40      - [ ] Unit tests for API client + review-state logic; `npm run build` clean
41
42      **L3 ? local end-to-end (owner's box, real match videos):**
43      - [ ] Full UI flow on localhost: upload ? job ? review/correct ? compile ? download
44      - [ ] `python run_all_tests.py` green; UI polling vs SQLite WAL sanity; WASB/CLI paths
untouched
45      - [ ] **GATE GL:** owner signs off local E2E ? only then deployment work begins
46
47      ### Phase D ? deployment (?2 sessions + ~1 hr owner)
48      - [ ] **D1 (owner):** Cloudflare account ? add khelsutra.guru ? GoDaddy nameserver
switch (HOSTING_PLAN $5) ? scoped API token ($6) ? Pages project connected to GitHub `khelsutra`
(build root `web/`)
49      - [ ] **D2 (Claude):** DNS records; `cloudflared` Windows service on the box ?
`api.khelsutra.guru`; Access apps + email-allowlist policies for `app` & `api`; SPA prod env
50      - [ ] **D3:** deployed smoke E2E (owner-only allowlist); go-live security checklist
(HOSTING_PLAN $7) all green
51      - [ ] **GATE GD:** checklist signed off
52
53      ### Phase R ? rollout
54      - [ ] Wave 1: 5 testers (C13/camera-pilot contacts) + feedback channel (WhatsApp group
or Google Form)
55      - [ ] 2-week retro ? punch list ? widen toward 25 ? beta planning (UI_PROPOSAL $4)
56      - Metrics: matches/week · % rallies corrected · median turnaround · verdict mix · week-2
tester retention
57
58      ## Working agreement
59      Claude (Cowork) designs, codes, writes tests + docs, and leaves a
`scratch/PRn_HANDOFF.md` per
60      work item. **Claude Code executes all git/PR mechanics** (branch, commit, push, PR,
conflict

```

21. user (2026-06-12T14:35:54.176Z)

Structured output provided successfully

22. assistant (2026-06-12T14:36:04.865Z)

Review complete. Findings returned via structured output: 4 medium issues (COGS mis-citation to the "quality table" when the figure comes from the 2026-06-10 audit; the PB Vision × PodPlay reference being unresolvable in-repo; the H5 "unprompted ??300" anchor contradicting the

script's own price ladder; PIVOT A/C triggers lacking numeric thresholds despite the pre-registration claim) and 4 low issues (Stupa missing from the flyer/answer-sheet incumbent lists; 75-150 session-cost arithmetic and match/session unit conflation; the PMF §6-risk-1 citation pointing at the wrong section; an H5 anchor gap for sub-100 numbers). No guardrail contradictions, broken links, or malformed tables were found.